Day 3 of 15 | Self-Check

Day 3 Self-Check Quiz — Probability, Correlation, Regression and the t-Test

Name: _____ Date: _____ Score: _____ / 15

Instructions: Write the letter of the best answer, or the value asked. You have 20 minutes. Passing score: 12/15 (80%).

Part I — Multiple Choice

1. ___ Out of 500 customers, 151 paid with GCash. P(GCash) is:

   A. 1.51 B. 0.151 C. 0.302 D. 3.31

2. ___ If P(Cash) = 0.56, then P(not Cash) is:

   A. 0.56 B. 0.44 C. 1.56 D. 0.06

3. ___ In a normal distribution, about what percent of values lie within **mean ± 2 SD**?

   A. 50% B. 68% C. 95% D. 100%

4. ___ A rice bag has z = −3.4. This means the bag is:

   A. Exactly average B. Slightly heavy C. Very unusually light D. 3.4 kg lighter

5. ___ r = −0.88 between price and packs sold means:

   A. Strong positive relationship B. Strong negative relationship C. No relationship D. Price causes 88% of sales

6. ___ "Ice-cream sales and drowning cases are correlated, so ice cream causes drowning." The error is:

   A. Wrong formula B. Correlation is not causation C. r must be negative D. Sample too large

7. ___ The regression line is $\hat{y} = -258 + 12.3x$ (x = temperature °C). The **slope** tells us:

   A. Sales when the temperature is 0 °C B. About 12.3 more drinks per extra °C C. The correlation D. The forecast error

8. ___ Which formula is written correctly?

   A. `=SLOPE(B2:B61,C2:C61)` with B = x, C = y B. `=SLOPE(C2:C61,B2:B61)` with B = x, C = y C. `=SLOPE(C2:C61)` D. `=SLOPE(x)`

9. ___ A t-test gives p = 0.0004. At α = 0.05 you conclude:

   A. No difference B. The difference is real (significant) C. The test failed D. Collect less data

10. ___ A t-test gives p = 0.27. The correct statement is:

    A. There is definitely no difference B. The difference is proven C. Not enough evidence; it could be chance D. p must be wrong

11. ___ While typing `=G4/G8`, pressing **F4** right after `G8` changes it to:

    A. `G$8` B. `$G$8` C. `$G8` D. `G8%`

12. ___ The shortcut that formats the selected cells as percent is:

    A. Ctrl + P B. Alt + = C. Ctrl + Shift + % D. Ctrl + D

Part II — Compute

13. Mean = 5.04 kg, SD = 0.04 kg. The z-score of a 5.10-kg bag = _____

14. Using $\hat{y} = -258 + 12.3x$, forecast the drinks sold at 35 °C = _____

Part III — Excel

15. Write the Excel formula that gives the two-tailed, unequal-variance t-test p-value comparing `B2:B91` and `C2:C91`: